



Circular Knitting

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Background

1.1 Context

This document summarises the modelling approach for circular knitting, representing a typical industrial process for jersey fabric production.

The purpose is to make the process transparent and reproducible.

1.2 Scope and functional unit

- **Functional unit:** 1 kg of knitted fabric.
- **Technology:** single and double jersey circular knitting.
- **Included:** lubricants, energy consumption, electricity, indirect energy allocation, capital goods and their end-of-life (EoL), and building infrastructure.
- **Excluded:** fibre and yarn manufacturing (modelled separately), textile/material waste (yarn losses and waste-yarn treatment), which is handled externally by the caller via the process-specific IOR, building construction costs beyond the amortised area share modelled in Section 3.6, packaging and logistics.
- **Background data:** ecoinvent 3.12, system model: cut-off by classification. Background processes are linked via generic market activities (GLO/RoW).

Methodological Overview

2.1 Yarn losses and lubricant - dependency on yarn fineness (dtex)

Concept

Circular knitting waste and lubricant use vary with yarn fineness.

Finer yarns require longer machine time, experience higher friction, and tend to break more often — leading to higher waste, lubricant use and, indirectly, energy demand.

The base definition follows the same logic as in flat knitting:

"Input quantity refers to the net yarn required to produce 1 kg of knitted fabric. Yarn losses during the knitting process (thread breaks, edge waste, trial runs) should be included. Typical material losses for knitting range between 1–5%, depending on yarn type and machine efficiency. If no primary data exist, use 2% as default."

(Laursen et al., 2007; van der Velden et al., 2014; Sandin et al., 2019 – Mistra)

Classification by yarn fineness (dtex)

Yarn category	Tex range	Approx. dtex	Description
Super fine	2 – 7.5 tex	< 75 dtex	Very fine yarns used for lightweight jerseys
Fine	7.5 – 16 tex	75 – 160 dtex	Common in high-quality apparel fabrics
Medium	16 – 40 tex	160 – 400 dtex	Typical for sportswear and medium-weight knits
Coarse	> 40 tex	> 400 dtex	Thicker yarns for heavy or technical fabrics

(Burkinshaw 2016; Banerjee 2014; Ray 2012; Australian Cotton 2022)

Adjustment logic

Parameter	Observation	Representative quote	Source
Yarn loss / waste	Finer yarns are more fragile and break more often during operation.	"Reducing the yarn linear density from 300 to 45 dtex significantly increased yarn breakage during machine operation."	Haque & Naebe (2024)
Lubricant	More yarn contact and friction increase lubricant demand.	"Yarns with lower dtex required increased lubricant pre-treatment due to higher friction and mechanical interaction."	Liu, Cano & Ardanuy (2025)
Indirect energy (HVAC, air, etc.)	Lower throughput → more energy per kg.	"Knitting finer yarns resulted in longer machine running time, thereby increasing energy use per kilogram of fabric."	Palamutcu (2016); Haque & Naebe (2024)
Capital allocation (machine)	Longer knitting time increases depreciation	"Lower tex yarns required longer processing time, raising capital	INTI (2021)

Parameter	Observation	Representative quote	Source
	per kg.	cost per kg of output."	

Parameter values applied

Yarn fineness	Yarn losses (%)	Lubricant (kg/kg fabric)
Super fine (<75 dtex)	5%	0.008
Fine (75–160 dtex)	4%	0.007
Medium (160–400 dtex)	3%	0.006
Coarse (>400 dtex)	2%	0.005

These values are used as a reference to adjust inventory parameters when the yarn fineness is known, using 0.005 kg/kg yarn as a lower-bound (Hasanbeigi & Price, 2015) and 0.008 kg/kg yarn as an upper-bound (Sandin et al., 2019, p. 46), with intermediate bands linearly stepped (0.007 and 0.006) in line with the qualitative evidence that lower dtex increases lubrication needs (Liu, Cano & Ardanuy, 2025). **Limitation:** the 0.007 and 0.006 values are not directly reported in the literature and should be treated as simplified interpolation defaults until factory-specific lubricant dosing data are available.

If not, the **medium range** (3% loss rate for IOR, 0.006 kg/kg lubricant) is applied by default. The loss-rate values parameterise the external IOR only; the foreground dataset does **not** include extra yarn input or textile-waste treatment exchanges.

53. Electricity Consumption

3.1 Concept

Electricity demand in knitting is mainly driven by **yarn fineness (dtex)**, machine speed, and fabric structure.

Finer yarns require longer knitting time and higher frictional work per kilogram of fabric, which increases energy use.

This dependency has been empirically demonstrated by van der Velden et al. (2014), who modelled electricity consumption as inversely proportional to yarn thickness. The electricity demand per kilogram of fabric, E (in kWh/kg), scales inversely with yarn fineness (dtex):

$$E \propto \frac{1}{\text{dtex}}$$

Sandin et al. (2019) proposed an alternative formulation based on Idemat/ITMF data, which is examined and re-fitted in §3.2.

3.2 Empirical foundation

For knitting, consumption values from Idemat (2012) and ITMF (2010) were interpolated by Sandin et al. (2019), producing a reference equation. Note: the original Sandin formulation ($E = 0.00303 + dtex/38.805$) is an **increasing function of dtex** and was found to be inconsistent with the physical expectation that finer yarns require more energy per kg. It was therefore discarded.

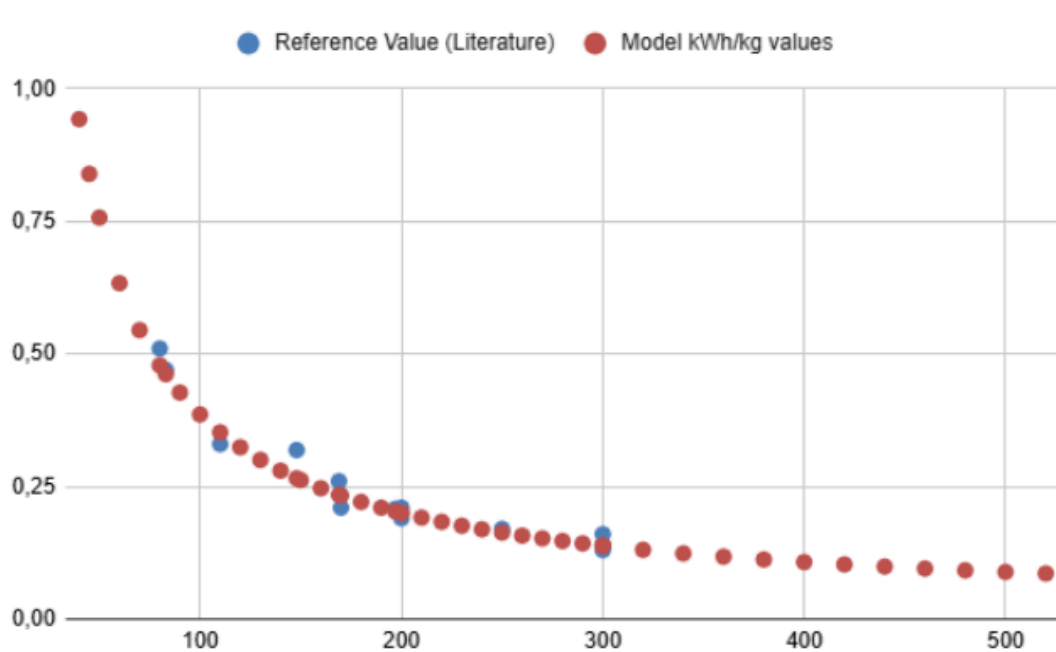
An expanded dataset of **13 anchor points** (80–300 dtex) from literature and factory records was used to derive an updated regression (ordinary least squares):

$$E_{\text{circular knitting}} = 0.01457 + \frac{37.119}{dtex} \text{ (kWh/kg fabric)}$$

with **RMSE = 0.0219 kWh/kg** and **R² = 0.967**.

This function decreases with dtex (consistent with physical expectation) and remains valid between **80 dtex and 400 dtex**. Outside this range, values are treated as extrapolations with increased uncertainty.

An original regression was performed using 13 literature and industry anchor points spanning 80–300 dtex. The table below provides the full dataset with primary source references.



dtex	E (kWh/kg)	Source
80	0.510	Van der Velden et al. (2014)
83	0.470	Idemat (2012)
110	0.330	Sandin et al. (2019), based on Cotton Inc. (2016)

dtex	E (kWh/kg)	Source
148	0.319	Koç & Çinçik (2010), derived
169	0.260	Klooster et al. (2024)
170	0.210	Sandin et al. (2019), based on Cotton Inc. (2016)
197	0.208	Koç & Çinçik (2010), derived
200	0.210	Van der Velden et al. (2014); ITMF (2010)
200	0.190	ITMF (2010)
250	0.170	Klooster et al. (2024)
300	0.140	Van der Velden et al. (2014)
300	0.130	Idemat (2012)
300	0.160	ITMF (2010)

All values represent machine electricity only (direct knitting energy, excluding indirect loads). Regression was performed using ordinary least squares (OLS) on the linearised form, which forces the physically correct inverse structure while allowing a non-zero intercept representing machine idle and auxiliary base loads.

3.3 Representative values

The model was then applied to generate reference values across typical yarn fineness ranges used in **circular knitting**.

For consistency, the corresponding NE, Nm and denier equivalents were also listed.

Representative Value (kWh/kg)	Circular Knitting dtex	Approx. denier	Nm	NE
0.76	40	45	250	148
0.54	60	63	167	98
0.48	80	72	125	74
0.39	100	90	100	59
0.32	120	108	83	49
0.26	150	135	67	40
0.23	170	153	59	35
0.22	180	162	55	33
0.20	200	180	50	30
0.18	230	207	43	27
0.16	250	225	40	24
0.15	270	245	37	22
0.14	300	270	33	20

Representative Value (kWh/kg)	Circular Knitting dtex	Approx. denier	Nm	NE
0.13	330	297	30	18
0.12	370	333	27	16
0.11	400	470	25	15
0.10	450	420	20	13

(Values rounded; derived from $E=0.01457+37.119/dtex$)

3.4 Interpretation

- Below 80 dtex, consumption rises sharply as machine efficiency drops.
- For coarser yarns (> 400 dtex), marginal energy reduction is minimal.

3.5 Application in LCI models

When primary data are unavailable, these reference values are used to populate the **electricity input** of knitting unit processes.

The selected value must correspond to the **yarn fineness (dtex)** of the fabric modelled.

For multi-yarn blends, a **mass-weighted dtex** is calculated prior to lookup.

4. Capital Goods, End-of-Life and Building Infrastructure

Shared methodology: The full modelling framework for capital goods, end-of-life allocation, and building infrastructure is documented in:

- [Capital Goods & End-of-Life \(Machinery\) in LCA](#)

- [Building Infrastructure in LCA](#)

We apply the same calculation logic used in the flat knitting process:

This section states only the circular-knitting-specific inputs.

4.1 Modelling approach

$$M_{\text{machine per kg fabric}} = \frac{M_{\text{machine}}}{Q_{\text{lifetime}}}$$

where

$$Q_{\text{lifetime}} = \text{productivity (kg/h)} \times \text{hours/day} \times \text{days/year} \times \text{years}$$

- **Machine mass** is taken from OEM brochures.

- **Productivity** is taken from manufacturer specifications (e.g., Mayer & Cie., Leadsfon).
- **Service life** parameters vary by scenario (see below), **we are using average.**

4.2 Lifetime parameters

Scenario	Years	Hours/day	Days/year	Productivity (kg/h)	Resulting allocation (kg/kg fabric)
Best case (BCS)	20	24	365	30	0.0005
Average case	15	16	300	30	0.0013
Worst case (WCS)	10	8	300	30	0.0039

These values correspond to the **mass of machine attributed per kilogram of fabric**.

They represent the “infrastructure weight” of the process and ensure consistency across product systems.

4.3 End-of-life allocation

Machine composition is assumed **2800kg**:

- **85% steel** → 2380 kg
- **15% other materials** → 420 kg

EoL per kg of fabric (average scenario):

Material	Allocation (kg/kg fabric)	Note
Steel	0.0011	Recycled or disposed depending on scenario
Other materials (Electric and Electronics)	0.0002	Non-ferrous components (motors, electronics)

4.4 Alignment with standards

- Complies with EF guidelines for capital goods inclusion.
- Mirrors ecoinvent's approach to infrastructure amortisation and decommissioning.
- Neutral and impact-method independent.

4.5 In ecoinvent format

For the “**market for building machine**” dataset:

Field	Entry
Amount	4.6E-07 unit/kg (average scenario)

Field	Entry
Unit	unit
Product	building machine
Dataset	market for building machine / GLO

(Derived from $0.0013 \text{ kg/kg fabric} \div 2800 \text{ kg machine weight}$. ecoinvent version: 3.12.)

4.6 Building Infrastructure

The circular knitting process model includes an optional building infrastructure (pavilion) share as a modular and auditable input. The building is represented by an ecoinvent-style construction dataset ("building, hall") whose reference flow is m^2 of industrial hall.

4.6.1 Allocation principle (area amortisation over lifetime output).

Building infrastructure is attributed to 1 kg of knitted fabric by amortising the allocated hall area over the lifetime production associated with the knitting cell:

$$\text{Building intensity (m}^2\text{/kg)} = \frac{A_{\text{alloc}} \text{ (m}^2\text{)}}{Q_{\text{lifetime}} \text{ (kg)}}$$

$$Q_{\text{lifetime}} = \text{productivity (kg/h)} \times \text{hours/day} \times \text{days/year} \times \text{years}$$

where A_{alloc} is the hall area allocated to the knitting cell (machine footprint + access + creel/yarn storage + maintenance space + internal logistics), and Q_{lifetime} is the total knitted output produced during the reference service life under the chosen operating regime.

4.6.2 Technology rationale.

Circular knitting is the highest-throughput knitting technology, operating continuously with minimal interruptions. Its compact cylindrical machine footprint combined with high production rates yields the lowest building intensity per kg of output among all knitting technologies considered.

4.6.3 Practical implementation.

In the circular knitting foreground unit process, we add an input: *building, hall* (unit = m^2) with amount = $1.00\text{E-}05 \text{ m}^2\text{/kg}$.

Value derived from the machine envelope (Terrot GmbH, $h = 2,550 \text{ mm}$, $b = 2,200 \text{ mm} \approx 5.6 \text{ m}^2$) $\times$ cell factor $\sim 6 = A_{\text{alloc}} \approx 35 \text{ m}^2$; $Q_{\text{lifetime}} = 2,160,000 \text{ kg}$ (average scenario); building intensity = $35 \div 2,160,000 = 1.62 \times 10^{-5} \text{ m}^2\text{/kg}$. The adopted default of $1.00 \times 10^{-5} \text{ m}^2\text{/kg}$ is conservative and consistent with the openLCA Nexus benchmark ($1.61 \times 10^{-5} \text{ m}^2\text{/kg}$).

4.6.4 End-of-life of the building.

The building dataset includes construction, maintenance and decommissioning within the background system model. No separate building EoL module is created in the foreground; only the amortised m^2/kg input is provided.

Default value (this study). In the absence of site-specific A_{alloc} and Q_{lifetime} , the following reference intensity is applied:

■ Circular knitting: **1.00E-05 m^2/kg**

This value is consistent with the benchmark reported in the openLCA Nexus organic cotton sweater case study (1.61E-05 m^2/kg for circular knitting). Physical plausibility was cross-checked using machine envelope data from Terrot GmbH single-jersey machines ($h = 2550$ mm, $b = 2200$ mm, envelope $\approx 5.6 \text{ m}^2$) with a conservative cell factor of $\sim 6\times$, yielding $A_{\text{alloc}} \approx 35 \text{ m}^2$ and a calculated intensity of $\sim 1.62\text{E-}05 \text{ m}^2/\text{kg}$ — same order of magnitude as the adopted default. Note: the openLCA Nexus value ($1.61 \times 10^{-5} \text{ m}^2/\text{kg}$) was developed by GreenDelta and is referenced here for cross-validation only. The adopted default was derived independently using our methodology described above.

5. Indirect Energy Allocation

■ **Shared methodology:** Full derivation and evidence base in [Indirect Energy in LCA — Shared Methodology \(Knitting\)](#).

Default value: 4.46 kWh/kg — derived from EcoBalyse upper estimate ($2.4 \text{ kWh/kg} = 35\%$ of total; ADEME, 2024): $2.4 \div 0.35 \times 0.65 = 4.46 \text{ kWh/kg}$. Temporary reference pending sub-metered factory data.

5.2 Evidence base

- Factory case showing **80%** of total electricity from HVAC and chillers (Sakti et al., 2021).
- Energy breakdown in spinning: **machines 78%, humidification 16%, lighting 3%, compressors 3%** (Hasanbeigi, 2010).
- Compressed air typically **10–40%** of total plant energy (Confederation of Indian Industry, n.d.).
- Cleaning accounts for **~40% of stoppage time** in knitting (Çelik et al., 2025).
- Weaving process benchmark: **machines 36.3%, compressors 29.4%, HVAC 27.1%**, total **5.06 kWh/kg** (Koç & Çinçik, 2010).

5.3 Default distribution (“mid-range knitting” set)

Subsystem	Share (%)	Comment
Knitting machines	35	Main mechanical load
Compressors	25	Pneumatic system for needle actuation
HVAC	30	Air conditioning and humidification
Lighting	6.5	Facility lighting
Cleaning / auxiliaries	3.5	Maintenance and automation units

This distribution is **identical for flat and circular knitting** until factory-level data allow refinement.

5.4 Total indirect energy

Extrapolating to include HVAC, air compression and auxiliaries, a consistent high-load total of **4.46 kWh/kg** is adopted.

Indirect energy = 4.46 kWh/kg fabric

Derivation from the EcoBalyse upper estimate for knitting machine electricity is 2.4 kWh/kg (ADEME, 2024), representing 35% of total plant electricity based on the subsystem distribution in Section 4.3. Total plant electricity = $2.4 \div 0.35 = 6.86$ kWh/kg; indirect share = $6.86 \times 0.65 =$ **4.46 kWh/kg**.

This figure is used for both **flat and circular knitting** as a temporary reference, pending real measurements.

It can later be split by the above percentages to isolate subsystem consumption.

Input–Output Ratio (IOR)

Definition: $\text{IOR} = 1 / (1 - L)$, where L is the yarn loss rate (as a fraction). For every 1 kg of finished knitted fabric, the upstream yarn input must be scaled by IOR.

How the loss rate is derived

Circular knitting loss rates are parameterised by yarn fineness class. Finer yarns break more frequently, increasing waste per kg of output. The IOR decreases with dtex, consistent with physical expectation. Values are consistent with Laursen et al. (2007), van der Velden et al. (2014), and Sandin et al. (2019).

IOR values by yarn fineness class

Yarn fineness class	dtex range	Loss (%)	IOR (kg/kg)	Input required (kg)	Loss per kg output (kg)
Super fine	< 75 dtex	5%	1.0526	1.0526	0.0526
Fine	75–160 dtex	4%	1.0417	1.0417	0.0417
Medium (default)	160–400 dtex	3%	1.0309	1.0309	0.0309
Coarse	> 400 dtex	2%	1.0204	1.0204	0.0204

Representative dataset values (selected dtex points):

Process specification	Loss (%)	IOR (kg/kg)
Circular Knitting, 70 dtex (63 den; 84/1 NE; 143 nm)	5%	1.0526
Circular Knitting, 80 dtex (72 den; 74/1 NE; 125 nm)	4%	1.0417
Circular Knitting, 120 dtex (108 den; 49/1 NE; 83 nm)	4%	1.0417
Circular Knitting, 150 dtex (135 den; 40/1 NE; 67 nm)	4%	1.0417
Circular Knitting, 170 dtex (153 den; 34/1 NE; 59 nm)	4%	1.0417
Circular Knitting, 200 dtex (180 den; 30/1 NE; 50 nm)	3%	1.0309
Circular Knitting, 250 dtex (225 den; 24/1 NE; 40 nm)	3%	1.0309
Circular Knitting, 300 dtex (270 den; 20/1 NE; 33 nm)	3%	1.0309
Circular Knitting, 370 dtex (333 den; 16/1 NE; 27 nm)	3%	1.0309
Circular Knitting, 400 dtex (470 den; 14/1 NE; 25 nm)	2%	1.0204

How to apply: multiply upstream yarn input by the IOR corresponding to the yarn fineness of the fabric being modelled. For multi-yarn blends, use the mass-weighted mean dtex to select the fineness class. Lost yarn is routed by the caller to for example `market for waste yarn and waste textile {Row}`. No inline yarn-loss or textile-waste exchange is included in the dataset itself.

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